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The Local Distance Network: A community consensus report on the measurement of the Hubble constant at ~1% precision

Authors: Community consensus (ISSI workshop Hubble-constant)

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ABSTRACT

Most precise direct measurement: H0 = 73.50 +-0.81 km/s/Mpc (1.09% relative uncertainty) covariance-weighted combination of local distance indicators. Compares to Planck+SPT+ACT: H0=67.24 +-0.35. Tension >7 sigma from that calibrated to CMB. Uses multiple orthogonal paths: MW+LMC/SMC Cepheids+TRGB+ SNIa+FP/SBF+MM etc. Validates against 42 Cepheid calibrators Riess et al. 2022b.

What was actually measured (numbers):

73.50 +-0.81 vs 67.24 +-0.35, tension 6-7 sigma, baseline uses Cepheids+TRGB HST+JWST

Devices / Protocol:

HST, JWST, Gaia parallaxes, Cepheids, TRGB, Mira, SNIa, Surface Brightness Fluctuations, Fundamental Plane, Tip of Red Giant Branch

Replication across teams / Independence:

Consensus combines >10 independent teams, two orthogonal paths O1/O2 agree, cross-checked vs Planck18, ACT2025, SPT-3G D1 2025, BBN+DESI BAO DR2 2025

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